

Alternating Current

SHORT NOTES

- ❖ The constant value of dc which produces same heat through a resistive element, as due to the alternating current, is known as root mean square value of A.C.
- ❖ 240 V ac is the rms value of ac voltage. The amplitude of this voltage is $V_M = 240 \times \sqrt{2} = 340$ volt.
- ❖ The power rating in ac circuit is the average power rating.
- ❖ Power consumed in a circuit is non negative.
- ❖ Phase relationships in a.c. circuits is best represented by phasor diagram. A phasor is a vector which rotates with the angular velocity ω . The magnitude of phasor is the peak value of voltage or current (V_o or I_o).
- ❖ In purely resistive AC circuit, voltage and current are in the same phase $V = V_o \sin \omega t$ and $I = I_o \sin \omega t$, where $I_o = \frac{V_o}{R}$.
- ❖ In purely resistive circuit, average power loss $= I_{rms}^2 \times R$, $I_{rms} = \frac{I_o}{\sqrt{2}}$, similarly $V_{rms} = \frac{V_o}{\sqrt{2}}$
- ❖ The only element which dissipates energy in ac circuit is resistor (R).
- ❖ In purely inductive circuit, inductive reactance $X_L = 2\pi fL = \omega L$ voltage is ahead of current by $\frac{\pi}{2}$, $V = V_o \sin \omega t$, $I = I_o \sin\left(\omega t - \frac{\pi}{2}\right)$, $I_o = \frac{V_o}{X_L}$. In this circuit, average power loss $= 0$.
- ❖ In purely inductive or capacitive circuit, $\cos \phi = 0 \Rightarrow \phi = \frac{\pi}{2}$.
Average power loss is zero. Although current is flowing in the circuit. Such a current is known as wattless current.
- ❖ In AC L-R circuit, total voltage $V = \sqrt{V_R^2 + V_L^2}$.
- ❖ In purely capacitive AC circuit, capacitive reactance $X_C = \frac{1}{2\pi fC} = \frac{1}{\omega C}$. The current leads the applied voltage by $\frac{\pi}{2}$ or 90° . $V = V_o \sin \omega t$, $I = I_o \sin\left(\omega t + \frac{\pi}{2}\right)$, $I_o = \frac{V_o}{X_C} = 2\pi fCV_o$. The average power loss per cycle is zero.
- ❖ In AC C-R circuit, total voltage $V = \sqrt{V_R^2 + V_C^2}$.
- ❖ For a given RLC circuit driven by voltage $V = V_o \sin \omega t$, the current is given by, $I = I_o \sin(\omega t + \phi)$ where $I_o = \frac{V_o}{\sqrt{R^2 + (X_L - X_C)^2}}$ and $\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$, impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$.
- ❖ The phase difference between voltage across L and voltage across capacitor C, is 180° . Thus $V_{LC} = V_L - V_C$.
- ❖ The voltage in series LCR AC circuit is given by $V = \sqrt{V_R^2 + (V_L - V_C)^2}$.
- ❖ The average power consumed $= V_{rms} \times I_{rms} \times \cos \phi$, where $\cos \phi$ is the power factor.
- ❖ In series LCR circuit, at resonance, $X_L = X_C$, the impedance Z is minimum and equal to R. In this case, the source frequency $\omega = \frac{1}{\sqrt{LC}}$ which equals resonant frequency.
- ❖ The quality factor $Q = \omega_0 \frac{L}{R} = \frac{1}{\omega_0 CR} = \frac{1}{R} \sqrt{\frac{L}{C}}$ is an indicator of “sharpness of resonance.”
- ❖ The power factor in a RLC circuit is a measure of how close the circuit is to consuming maximum power.
- ❖ A step up transformer converts low AC voltage to high AC voltage but reduces the current.
- ❖ A step down transformer converts high ac voltage to a low AC voltage but increases the currents accordingly.
- ❖ In transformer, the primary and secondary voltage are given by $V_s = \left(\frac{N_s}{N_p}\right) V_p$ and the current are given by $I_s = \left(\frac{N_p}{N_s}\right) I_p$.
In step up transformer, $N_s > N_p$ and step down transformer $N_s < N_p$.
- ❖ A transformer does not violate the law of conservation of energy. A step up transformer changes low voltage to a high voltage but reduces the current in the same proportion.